**9.** **In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \qquad x > 0$$

Find the $x$ coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

**(6)**